Aaron Lin

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EDUCATION

Boston University Boston, MA
B.S. Mechanical Engineering, Aerospace Concentration, Materials Science & Engineering Minor Expected May 2028
GPA: 3.76/4.0, Dean's List

SKILLS

Design & Simulation: SolidWorks (Simulation, HSMWorks), ANSYS (Fluent, Mechanical), OpenRocket, Onshape
Programming & Test: Python, MATLAB, C++, optical emission spectroscopy
Fabrication: FDM and resin 3D printing, CNC machining, manual lathe and mill, soldering

EXPERIENCE & PROJECTS

Project Engineer Sep 2024 – Present
Boston University Rocket Propulsion Group Boston, MA

- Designing a fairing for ARGO's external plumbing and evaluating candidate shapes at a Mach 2.81 design condition using ANSYS Fluent and OpenRocket.
- Designed a converging–diverging nozzle in SolidWorks and produced part and assembly drawings. Calculated 35.0 N of thrust under optimal conditions and a minimum factor of safety of 2.12 in pressure-only FEA at 80 psi.
- Resin-printed the nozzle and helped integrate it with the chamber, ignition system, and data acquisition for a team static fire.

Mechanical Engineering Intern Feb 2026 – Jun 2026
Culturon (plasma-activated coatings) Sydney, Australia

- Ran optical emission spectroscopy experiments and wrote repeatable procedures to check plasma consistency and detect elemental contaminants.
- Built a Python workflow to find emission peaks and compare them with the NIST atomic spectra database.
- Benchmarked the plasma chamber against a reference chamber and documented characteristic emission lines.

CAD and Mechanical Layout Lead Sep 2026 – Oct 2026
Cartesian Fishing Game (ME360) Boston, MA

- Led SolidWorks design for a joystick-controlled gantry with three belt-driven stages and a motorized magnetic pickup.
- Designed 3D-printed motor and pulley mounts. The team demonstrated X–Y positioning, magnetic pickup and release, and all three timed rounds.

Firmware and Integration Contributor Sep 2025 – Dec 2025
Autonomous Desk Cleaning Robot Boston, MA

- Wrote C++ firmware for a robot that cleaned a 4 ft by 4 ft desk, started with a hand gesture, and turned when sensors detected an edge.
- Helped refine the suction system. The team placed in the top four of about 20, and the robot collected 72 of 95 rice grains in the final trial.